Monoethylene Glycol (MEG)Market Analysis 2019–2024

Global Supply-Demand Dynamics, Price Trends & Competitive Landscape

EO/EG Competitive Intelligence: Dow • SABIC • Shell

Prepared for: Petrochemical Strategy & Advisory | Date**:** May 2026

Table of Contents

---

1. Executive Summary
2. MEG Market Overview
3. Global MEG Supply-Demand Analysis (2019–2024)
4. MEG Price Trends (2019–2024)
5. EO/EG Competitive Landscape: Dow vs. SABIC vs. Shell
6. U.S. EG/EO Unit Insights
7. Key Market Themes & Strategic Outlook
8. Appendix

1. Executive Summary

The global monoethylene glycol (MEG) market has undergone a period of profound structural transformation between 2019 and 2024. What was once a relatively stable commodity market dominated by Middle Eastern exporters has been reshaped by a confluence of capacity expansions, geopolitical shifts, pandemic-era disruptions, and strategic repositioning by major producers.

Key Findings

- **Market Scale:** Global MEG demand reached approximately **28 million tonnes** in 2022 and has been growing at a compound annual growth rate (CAGR) of approximately **3.2%**, driven primarily by polyester fiber, PET resin, and packaging applications.
- **Capacity Expansion Wave:** Global ethylene glycol production capacity surged from approximately 38 million tonnes in 2019 to nearly 57 million tonnes by 2022, propelled by China's private refining sector, syngas-based MEG routes, and North American shale-gas projects.
- **China's Self-Sufficiency:** China's domestic MEG capacity has expanded dramatically through both conventional ethylene-based and coal/syngas-to-MEG pathways, significantly reducing the country's dependence on imports and reshaping global trade flows.
- **Price Volatility:** MEG prices experienced extreme swings—collapsing in 2020 during the COVID-19 pandemic, spiking in 2021–2022 amid supply-chain disruptions and energy cost inflation, then normalizing under growing oversupply pressure in 2023–2024.
- **Competitive Landscape Shifts:** Dow is strategically pivoting away from commodity MEG toward higher-margin purified EO derivatives, while SABIC continues to leverage low-cost Middle Eastern feedstocks, and Shell maintains a dual competitive moat as both the world's leading EO/EG technology licensor and an integrated producer.
- **U.S. Trade Disruptions:** Rising tariff barriers on U.S. MEG exports to China (2024 onward) have forced American producers to redirect volumes to India, Indonesia, and other Asian markets.

◆ **Executive Takeaway**

The MEG market has entered a structurally oversupplied environment. While underlying demand growth remains healthy at ~3.2% CAGR, the pace of capacity additions—particularly in China—has outstripped consumption growth. Producers without a clear cost, technology, or integration advantage face intensifying margin pressure through the remainder of the decade.

2. MEG Market Overview

2.1 What is Monoethylene Glycol?

Monoethylene glycol (MEG), with the chemical formula $C_2H_6O_2$ (or $HOCH_2CH_2OH$), is a colorless, odorless, viscous liquid that is completely miscible with water. It is the simplest member of the glycol family and serves as one of the most widely consumed petrochemical intermediates globally.

Property	Detail
Molecular Formula	$C_2H_6O_2$
Molecular Weight	62.07 g/mol
Boiling Point	197.3 °C
Density	1.113 g/cm³
Key Characteristics	Hygroscopic, low freezing point, high thermal stability, excellent solvent properties
CAS Number	107-21-1

2.2 Primary Applications & Demand Segmentation

MEG's versatility underpins its broad application base. The two dominant end uses—polyester fiber and PET resin—together account for approximately 80% or more of global MEG consumption.

Application	Share of Global Demand (est.)	Description
Polyester Fiber / Yarn	~58%	Reacted with purified terephthalic acid (PTA) to produce polyester staple fiber (PSF) and polyester filament yarn (PFY) for textiles
PET Resin	~24%	Key feedstock for polyethylene terephthalate (PET) bottles, food packaging, and containers
Polyester Film	~5%	Used in electrical insulation, packaging films, and industrial applications
Antifreeze / Coolants	~8%	Automotive engine coolants, HVAC heat-transfer fluids, aircraft deicing
Other	~5%	Unsaturated polyester resins, brake fluids, adhesives, paints, solvents, plasticizers

2.3 End-Use Industry Segmentation

- **Textiles & Apparel:** The single largest demand driver, centered in Asia (China, India, Indonesia, Vietnam). Polyester fiber production tracks consumer textile demand and fast-fashion cycles.
- **Packaging:** PET bottles and food-grade packaging represent a steadily growing segment, supported by beverage consumption growth and recycling/rPET initiatives.
- **Automotive:** Coolants and antifreeze applications; steady demand linked to vehicle fleet size and seasonal patterns, with growing EV thermal-management demand.
- **Construction:** Unsaturated polyester resins used in construction composites, adhesives, and coatings.

2.4 Manufacturing Routes

MEG is produced through three principal commercial pathways, each with distinct feedstock, cost, and geographic profiles:

Route	Process Description	Key Regions	Economics
Ethylene Oxide Hydration (Conventional)	Ethylene is oxidized to ethylene oxide (EO) using a silver catalyst, then EO is hydrated thermally to produce MEG (with DEG/TEG co-products)	Middle East, North America, Europe, Northeast Asia	Dominant route; economics driven by ethylene/ethane feedstock cost
Shell OMEGA Catalytic Process	EO is catalytically converted to MEG via ethylene carbonate intermediate, achieving >99% MEG selectivity and minimizing DEG/TEG by-products	Licensed globally; Shell plants in U.S. and Europe	Higher MEG yield per unit of EO; lower by-product generation
Coal / Syngas-to-MEG	Coal is gasified to produce syngas (CO + H_2), which is converted through dimethyl oxalate (DMO) or other intermediates to MEG	China (dominant)	Lower feedstock cost where coal is cheap; higher capex and carbon intensity

Note: Bio-based MEG routes (from bioethanol or sugar-derived feedstocks) exist but remain commercially marginal at <1% of global supply.

◆ Key Takeaway

MEG demand is fundamentally tied to polyester and PET production, which together represent over 80% of consumption. China's coal-to-MEG pathway has introduced a structurally different cost curve into the market, enabling domestic Chinese producers to compete with traditional ethylene-based producers even when crude oil prices are moderate.

3. Global MEG Supply-Demand Analysis (2019–2024)

3.1 Global Capacity, Production & Demand Summary

The period from 2019 to 2024 has been characterized by an unprecedented wave of MEG capacity additions. Global production capacity expanded from approximately 38 million tonnes in 2019 to an estimated 62–65 million tonnes by 2024, representing a compound growth rate of roughly 10–11% per annum in capacity—far outstripping the ~3.2% annual growth in demand.

Year	Global Capacity (MM MT)	Global Production (MM MT)	Global Demand (MM MT)	Capacity Utilization	Surplus / (Deficit)
2019	~38.0	~27.5	~26.8	72%	+0.7
2020	~42.0	~26.0	~25.5	62%	+0.5
2021	~48.0	~28.5	~27.2	59%	+1.3
2022	~57.0	~29.5	~28.0	52%	+1.5
2023	~60.0	~30.5	~29.0	51%	+1.5
2024E	~63.0	~31.5	~29.9	50%	+1.6

Sources: Statista, industry estimates. Capacity figures include nameplate capacity for all ethylene glycol (MEG + DEG + TEG); demand figures represent MEG-equivalent consumption. "E" denotes estimate.

3.2 Regional Capacity Distribution

The geographic distribution of MEG capacity has shifted markedly. China's share of global capacity has risen rapidly since 2019, driven by both mega-refinery integrated complexes (e.g., Hengli Petrochemical, Zhejiang Petrochemical, Shenghong Petrochemical) and coal/syngas-to-MEG plants in inland provinces.

Region	2019 Capacity (MM MT)	2019 Share	2024E Capacity (MM MT)	2024E Share	Key Drivers
China	~8.5	22%	~25.0	40%	Mega-refineries, coal/syngas-to-MEG plants, self-sufficiency policy
Middle East	~11.0	29%	~13.5	21%	SABIC Jubail 700 KTA new plant (2021); low-cost ethane feedstock
North America	~6.5	17%	~8.5	13%	Shale-gas advantage; MEGlobal Oyster Creek (2019); Dow/Shell Gulf Coast
Rest of Asia-Pacific	~7.5	20%	~10.0	16%	India, Taiwan, South Korea, Japan; selective expansions
Europe	~3.0	8%	~3.0	5%	Largely stable; Shell Moerdijk (Netherlands) is the key hub
Other	~1.5	4%	~3.0	5%	Latin America, Africa; limited but growing
Global Total	**~38.0**	**100%**	**~63.0**	**100%**	**+66% capacity growth over 5 years**

Sources: Statista (global capacity 2022: ~57 MM MT), CCFGroup, ChemAnalyst, ICIS Supply & Demand Database, industry estimates.

3.3 Key Supply-Side Dynamics

Phase 1: Saudi Supply Dominance (Pre-2020)

Prior to 2020, Saudi Arabia (primarily through SABIC and its affiliates) was the dominant global MEG exporter, supplying the rapidly growing Chinese market with large volumes of competitively priced, ethane-based MEG. The Middle East's inherent feedstock cost advantage—utilizing abundant, low-cost ethane—positioned it as the world's marginal cost leader.

Phase 2: North American Capacity Surge (2019–2021)

The U.S. shale-gas revolution enabled a wave of new EO/EG capacity on the Gulf Coast. MEGlobal's world-scale EG facility at Oyster Creek, Texas commenced operations in 2019, co-located with Dow's infrastructure. Shell's Geismar complex and Dow's existing Gulf Coast sites expanded output. North American MEG exports to Asia surged during this period.

Phase 3: China's Mega Build-Out (Post-2021)

Starting in 2021, China commissioned an extraordinary volume of new MEG capacity through two parallel pathways: (1) integrated mega-refinery complexes incorporating world-scale EO/EG units, and (2) coal/syngas-to-MEG plants in coal-rich provinces. By 2024, coal-to-ethylene glycol accounted for approximately 34% of China's total EG output. China's self-sufficiency rate climbed sharply, reducing its once-voracious appetite for imports and fundamentally reshaping global trade flows.

3.4 Demand-Side Analysis

- **Asia-Pacific dominance:** The Asia-Pacific region accounts for approximately 65–70% of global MEG consumption, with China alone representing ~45% or more of global demand.
- **Textile/polyester recovery:** Post-COVID demand recovery in 2021–2022 was robust, driven by restocking and pent-up consumer spending on apparel and home textiles.
- **PET packaging:** Steady growth in beverage packaging, with increasing emphasis on recycled PET (rPET) content not materially displacing virgin MEG demand to date.
- **Automotive coolants:** Stable demand, with marginal growth from EV thermal-management fluid requirements.

3.5 Structural Oversupply Concerns

The most critical structural issue facing the global MEG market is the widening gap between nameplate capacity and actual demand. Global capacity utilization rates have declined from approximately 72% in 2019 to an estimated 50% by 2024. While operational discipline (maintenance shutdowns, idling of high-cost units, and strategic pivots like Dow's Seadrift EG closure) has prevented a full-scale collapse, the overhang of available capacity exerts persistent downward pressure on margins and utilization across all regions.

◆ **Key Takeaway**

Global MEG capacity has nearly doubled since 2019 while demand has grown by only ~12%. China's build-out has been the single largest driver, with the country's share of global capacity rising from 22% to 40%. This structural oversupply is now the defining feature of the global MEG market and is triggering rationalization of higher-cost assets in North America and Asia.

4. MEG Price Trends (2019–2024)

4.1 Regional Price Summary

MEG prices have exhibited significant volatility over the 2019–2024 period, driven by an interplay of feedstock cost movements, demand shocks, supply-chain disruptions, and the structural shift toward oversupply. The table below presents representative annual average prices across key regional benchmarks.

Year	US FOB Houston (USD/MT)	China CFR (USD/MT)	Asia CFR SE Asia (USD/MT)	Europe CIF NWE (USD/MT)
2019	580–640	550–620	560–630	620–700
2020	350–440	380–470	370–450	400–500
2021	550–750	600–850	580–800	650–900
2022	600–800	580–780	590–760	700–950
2023	450–550	420–530	430–540	500–620
2024E	470–540	440–520	450–530	520–620

Sources: Industry Sources. Ranges represent annual low–high averages; spot prices may have deviated further. "E" denotes estimate based on available data through Q4 2024.

4.2 Year-by-Year Price Analysis

2019: New Capacity Pressure on Moderate Prices

MEG prices in 2019 were moderate but under pressure from new capacity coming online, including MEGlobal's Oyster Creek facility and several Chinese plants. Asian CFR prices averaged in the $550–$630/MT range. The U.S.–China trade tensions introduced early uncertainty around tariff impacts on MEG trade flows.

2020: COVID-19 Demand Shock and Price Collapse

The pandemic triggered a severe demand contraction across textiles, automotive, and packaging sectors. U.S. FOB Houston prices dropped to the $350–$440/MT range—among the lowest levels in over

a decade. Chinese import demand weakened as downstream polyester operations curtailed. Producers deferred maintenance and idle capacity in response.

2021: Recovery, Supply-Chain Disruptions & Volatile Spikes

A sharp demand rebound coincided with global supply-chain disruptions, container freight rate spikes, and weather-related force majeures (including Winter Storm Uri on the U.S. Gulf Coast in February 2021). U.S. FOB Houston prices swung from ~$550/MT to above $750/MT. Asian prices exhibited similar volatility.

2022: Energy Crisis and Elevated Feedstock Costs

Russia's invasion of Ukraine and the resulting European energy crisis drove ethylene and naphtha feedstock costs sharply higher. European CIF MEG prices reached as high as $900–$950/MT. U.S. producers benefited from relatively lower ethane costs but still saw elevated price levels. Global demand remained healthy at ~28 million tonnes.

2023: Normalization and Growing Oversupply Pressure

Energy costs retreated, and the full weight of new Chinese capacity began to impact the market. Prices normalized to the $420–$550/MT range across regions. Chinese import volumes declined as domestic production rose, reducing the "pull" on Middle Eastern and North American exports.

2024: Mixed Regional Signals

Q1 2024 saw a bullish trend across most regions, driven by rising ethylene feedstock prices and robust demand from textile, automotive, and packaging sectors. However, Q2 2024 turned mixed, with North American and Kuwaiti prices turning bearish while Saudi and Asia-Pacific markets maintained a bullish bias due to supply discipline and firm downstream demand.

4.3 2025 Reference Points

For context, available 2025 data shows continued downward pressure: Q1 2025 U.S. MEG prices were approximately **$498/MT FOB Houston** (a modest +0.8% QoQ increase), before declining to approximately **$444/MT** in Q2 2025 (–10.8% QoQ), reflecting soft downstream demand and high inventory levels.

4.4 Key Price Drivers

Driver	Impact Mechanism	2019–2024 Significance
Ethylene / Crude Oil Feedstock Costs	Ethylene is the primary feedstock for conventional EO/EG; crude oil prices drive naphtha-based ethylene costs	Very High — the 2022 energy crisis drove prices to multi-year highs
Chinese Import Demand Shifts	As China's domestic capacity rises, import demand falls, removing price support for exporters	High — progressive impact from 2021 onward
Global Supply-Chain & Logistics	Container freight rates, port congestion, and shipping disruptions affect delivered costs	High in 2021–2022; normalized by 2023
Trade Tariffs & Geopolitics	U.S.–China tariffs affect bilateral trade flows; Middle East conflicts affect freight insurance	Moderate to High — tariffs redirected U.S. exports from 2024
Capacity Utilization / Oversupply	Low utilization rates exert structural downward pressure on spot prices	Growing significance from 2022 onward
Seasonal / Weather Events	Gulf Coast hurricanes, winter storms disrupt U.S. production; Chinese holiday demand cycles	Episodic but impactful (Uri 2021, Enzo 2025)

◆ **Key Takeaway**

MEG prices have trended structurally lower from the 2021–2022 spike, with the market settling into a $420–$550/MT range in most regions by 2023–2024. The growing oversupply, combined with declining Chinese import dependency, suggests that MEG producers will face persistent margin compression unless high-cost capacity is rationalized or demand growth accelerates beyond the ~3.2% CAGR baseline.

5. EO/EG Competitive Landscape: Dow vs. SABIC vs. Shell

The global EO/EG competitive landscape is shaped by three distinct strategic archetypes represented by Dow, SABIC, and Shell. Each company brings different strengths—integration and scale, feedstock cost advantage, or technology licensing moats—to the market. The following analysis compares their positions.

5.1 Comparative Overview

Dimension	Dow Chemical	SABIC	Shell
Headquarters	Midland, Michigan, USA	Riyadh, Saudi Arabia	London, UK / The Hague, Netherlands
Global EO/EG Capacity (est.)	~2,500–3,000+ KTA (incl. 42.5% EQUATE/MEGlobal stake)	~3,500–4,000+ KTA (Jubail complex + affiliates); world's largest EG producer	~1,500–2,000 KTA (own operations); +40% of world EO in Shell-licensed plants
Key Plant Locations	Freeport, TX; Seadrift, TX; Plaquemine, LA; MEGlobal Oyster Creek, TX; EQUATE (Kuwait)	Jubail Industrial City, Saudi Arabia (multiple units incl. new 700 KTA plant); affiliates in Asia	Geismar, LA; Norco, LA; Deer Park, TX; Moerdijk, Netherlands
Technology	Conventional EO hydration; Scientific Design (SD) catalyst/process technology (via SABIC's SD ownership)	SD-licensed EO/EG processes; Jubail CCU integration; SABIC's CO_2 recovery technology	Proprietary Shell MASTER & OMEGA processes; Shell HS/HP EO catalysts; world's leading EO/EG process licensor
Feedstock Advantage	U.S. shale-gas ethane (low-cost); Gulf Coast integration for ethylene supply	Middle East ethane (lowest global cost); advantaged gas price under Saudi allocation	U.S. shale-gas ethane (Geismar/Norco); European naphtha (Moerdijk)
Strategic Positioning	Pivoting from commodity MEG to higher-margin purified EO derivatives (ethanolamines, glycol ethers); leveraging	Scale leader in commodity EG; leveraging lowest-cost feedstock position; investing in decarbonization (CCU/CCUS) for	Dual moat: technology licensor + catalyst supplier + integrated producer; earns licensing/catalyst revenue regardless of MEG price cycle

Dimension	Dow Chemical	SABIC	Shell
	MEGlobal partnership for EG volumes	competitive differentiation	
Recent Strategic Moves	Seadrift EG unit indefinitely shut (March 2025); focus on purified EO; expanded ethylene supply to MEGlobal (+100 KTA, October 2025)	New Jubail 700 KTA EG plant started up (2021); SABIC/Linde/SD MoU to decarbonize EO/EG process (2024)	Continued catalyst innovation (~90% average selectivity over 3+ year life); AO4 alpha olefins expansion at Geismar (2018)
Key Financial Metric	Dow Inc. 2024 revenue: ~$43 billion; ~36,000 employees; operations in 30 countries	2025 total production: 55.5 MM MT; net income: $0.55 billion; total assets: $65.14 billion; operations in 44 countries with 60 plants	Shell Geismar is world's largest alpha olefins producer (1.3+ MM TPA); over 50% of global EO uses Shell catalysts

5.2 Dow Chemical — Detailed Profile

Dow is a major EO/EG producer on the U.S. Gulf Coast with a complex web of owned and affiliated assets. Its strategic direction over the past several years has been shifting away from commodity MEG toward higher-value purified EO derivatives.

Key assets and developments:

- **Freeport, TX:** Integrated EO/EG complex; remains Dow's core EO/EG production hub on the Gulf Coast.
- **Seadrift, TX:** EG unit with 300 KTA capacity was **indefinitely shut down in March 2025** following an electrical failure caused by Winter Storm Enzo in late January 2025. Dow elected not to restart the EG unit after the planned turnaround. The EO unit at Seadrift restarted in May 2025, with Dow pivoting to producing purified EO for downstream derivatives including ethanolamines and glycol ethers.
- **Plaquemine, LA:** Glycol II unit experienced a **major explosion and ethylene oxide release in July 2023**. A U.S. Chemical Safety Board (CSB) investigation found that foreign debris caused a rupture disc failure. The incident underscored operational risk at aging EO/EG facilities.

- **MEGlobal / EQUATE Partnership:** Dow holds a **42.5% ownership stake in EQUATE Petrochemical Company**. MEGlobal, a subsidiary of EQUATE, is the world's second-largest EG producer. MEGlobal started up its world-scale EG manufacturing facility at Oyster Creek, TX in 2019, co-located with Dow infrastructure.
- **Ethylene Supply Expansion:** In October 2025, Dow finalized an agreement to supply an additional equivalent of **100 KTA of ethylene** from its Gulf Coast operations to MEGlobal's Oyster Creek facility.

> ◆ **Strategic Insight — Dow**
>
> Dow's indefinite shutdown of the Seadrift EG unit is the clearest signal yet of its strategic pivot away from commodity MEG. By focusing on purified EO derivatives and leveraging the MEGlobal partnership for EG volumes, Dow is repositioning toward higher-margin products while reducing direct exposure to oversupplied glycol markets.

5.3 SABIC — Detailed Profile

SABIC is the world's largest (or among the largest) producer of ethylene glycol, anchored by its massive operations in Jubail Industrial City, Saudi Arabia. The company's primary competitive advantage is structural: access to the world's lowest-cost ethane feedstock under Saudi Arabia's advantaged gas allocation framework.

Key assets and developments:

- **Jubail Industrial City:** Multiple EO/EG production units, including its affiliate Jubail United Petrochemical Company (United). The new **Ethylene Glycol Plant – 3 at United started up in November 2021** with an estimated annual production capacity of **700,000 metric tonnes** of monoethylene glycol.
- **Global Footprint:** Headquartered in Riyadh; operates in **44 countries** with **60 manufacturing plants**; total production of **55.5 million metric tonnes** (2025).
- **Market Leadership:** Market leader in ethylene glycol, methanol, MTBE, and polyethylene. SABIC's EG operations are among the most cost-advantaged globally.

- **NOVA Chemicals:** SABIC holds an **11.5% ownership stake** in the NOVA Chemicals Geismar olefins facility (jointly with NOVA at 88.5%).
- **Decarbonization:** SABIC, Linde Engineering, and Scientific Design signed an MoU (2024) to jointly explore decarbonizing the EO/EG production process. SABIC's Jubail CCU plant already captures up to 500,000 tonnes of CO_2 annually.
- **Financials:** Net income $0.55 billion in 2025; total assets $65.14 billion.

◆ Strategic Insight — SABIC

SABIC's enduring competitive moat is its access to the lowest-cost ethane feedstock in the world. Even in an oversupplied market, SABIC's Jubail operations remain profitable at price levels that push higher-cost producers into negative margins. The addition of 700 KTA capacity in 2021 reinforced its position as the scale leader in commodity EG production.

5.4 Shell — Detailed Profile

Shell occupies a unique position in the EO/EG value chain as simultaneously the world's leading process technology licensor, the dominant EO catalyst producer, and a significant integrated producer. This "triple role" provides Shell with diversified revenue streams and competitive intelligence across the entire global EO/EG industry.

Key assets and developments:

- **Technology Licensing:** Shell has issued licenses for **89 EO/EG plants worldwide**, 53 of which are still operating. Nearly **40% of the world's EO is produced in Shell-licensed and designed plants**. Over **50% of the world's current EO production** is manufactured using Shell EO catalysts.
- **Two Licensed Processes:**
  - **Shell MASTER:** Catalytic conversion of ethylene to EO + thermal conversion of EO to EG. MEG yield of 90–92%. Two variants: MASTER^{HS} (optimized for High Selectivity catalysts) and MASTER^{HP} (lower capex, higher work rates).
  - **Shell OMEGA:** Catalytic conversion of both ethylene to EO and EO to EG (via ethylene carbonate intermediate). Achieves >99% MEG selectivity, virtually eliminating DEG/TEG by-products.
- **Catalyst Innovation:** Pioneer in high-selectivity EO catalysts. Selectivity improved from **68–70%** in early days to **80%** (1971 breakthrough), then to **86%+** with the introduction of HS catalysts (1986 breakthrough). Current catalysts enable approximately **90% average selectivity over 3+ year catalyst life**.
- **Geismar, LA:** Produces EO, EG, alpha olefins, alcohols, and ethoxylates. World's largest alpha olefins producer with **1.3+ million TPA** capacity after AO4 expansion in 2018.
- **Norco, LA & Deer Park, TX:** Provide ethylene feedstock to the Geismar complex.
- **Moerdijk, Netherlands:** Major European EO/EG hub.

◆ Strategic Insight — Shell

Shell's dual role as technology licensor/catalyst supplier and integrated producer creates a uniquely defensible competitive moat. Shell earns licensing fees and catalyst sales revenue from competitors' plants worldwide, providing annuity-like income that is less sensitive to MEG price cycles. The company's continuous catalyst innovation (from 68% to ~90% selectivity) has saved the industry billions in ethylene feedstock costs while reinforcing Shell's position as the indispensable technology partner.

---

6. U.S. EG/EO Unit Insights

6.1 Major U.S. EG/EO Production Units

Facility / Owner	Location	Products	Capacity (KTA)	Status / Notes
Dow — Freeport Complex	Freeport, TX	EO, EG (MEG/DEG)	Large-scale integrated	Core Dow EO/EG hub; operational. Multiple units.
Dow — Seadrift	Seadrift, TX	EO, EG	~300 (EG)	**EG unit indefinitely shut down** (March 2025) after electrical failure from Winter Storm Enzo. EO unit restarted May 2025 for purified EO production.
Dow — Plaquemine Glycol II	Plaquemine, LA	EO, EG	Medium-scale	Major explosion & EtO release **July 2023**; CSB investigation found foreign debris caused rupture disc failure.
MEGlobal — Oyster Creek	Oyster Creek, TX	EG (MEG/DEG)	World-scale (~750+)	Started up 2019; co-located with Dow; EQUATE subsidiary (Dow 42.5% stake). Expanded ethylene supply agreement Oct 2025 (+100 KTA equiv.).
Shell — Geismar	Geismar, LA	EO, EG, alpha olefins, alcohols, ethoxylates	EO/EG: significant; AO: 1,300+ TPA	World's largest alpha olefins producer. AO4 expansion completed 2018. Uses Shell

Facility / Owner	Location	Products	Capacity (KTA)	Status / Notes
				MASTER/OMEGA processes and Shell EO catalysts.
Shell — Norco	Norco, LA	Ethylene, propylene (feedstock supply)	Major ethylene cracker	Supplies ethylene to Geismar complex via pipeline.
Shell — Deer Park	Deer Park, TX	Ethylene, petrochemicals	Major complex	Provides additional ethylene feedstock to Geismar operations.
NOVA / SABIC — Geismar	Geismar, LA	Ethylene (cracker)	Major ethylene cracker	Joint venture: NOVA Chemicals 88.5%, SABIC 11.5%. Supplies ethylene to the Gulf Coast market.

6.2 North American Capacity Surge: The Shale-Gas Advantage (2019–2021)

The U.S. shale-gas revolution transformed the economics of North American petrochemical production. Abundant, low-cost ethane enabled Gulf Coast EO/EG producers to achieve feedstock cost positions competitive with Middle Eastern producers. Key capacity additions during this period included:

- **MEGlobal Oyster Creek (2019):** The world-scale EG plant commenced operations, adding significant new export-oriented capacity to the U.S. Gulf Coast. Co-location with Dow's infrastructure provides operational synergies and feedstock flexibility.
- **Existing Plant Expansions:** Debottlenecking and optimization at Dow's Freeport and Shell's Geismar complexes increased effective production capacity.
- **Export Growth:** North American MEG exports to China surged between 2019 and 2021, leveraging competitive U.S. ethane economics and strong Chinese import demand.

6.3 Trade Flow Shifts: The Tariff Barrier Challenge (2024 Onward)

The escalation of U.S.–China trade tariffs from 2024 onward has created a significant headwind for American MEG exporters. China—historically the primary destination for U.S. MEG exports—imposed steep tariffs that rendered U.S.-origin MEG uncompetitive relative to Middle Eastern and domestic Chinese supply. In response:

- **U.S. MEG exports are being redirected** from China to **India, Indonesia, and other Southeast Asian markets**, where tariff barriers are lower and MEG demand is growing.
- **Trade route complexity** has increased, with longer shipping distances to new markets partially offsetting the feedstock cost advantage.
- **China's MEG import structure** is shifting toward greater reliance on Middle Eastern, South Korean, and domestic sources at the expense of North American cargoes.
- **U.S. producers** face the dual challenge of losing their largest export market while contending with domestic oversupply pressure.

> ⚠ **Risk Factor**
>
> The combination of Dow's Seadrift EG shutdown, the Plaquemine Glycol II incident, and tariff-driven export redirection signals a challenging operating environment for U.S. Gulf Coast EG producers. Producers without structural cost advantages (shale-gas feedstock) or integration benefits (co-located EO derivatives) face the greatest exposure to margin compression.

7. Key Market Themes & Strategic Outlook

7.1 Defining Themes Shaping the MEG Market

#	Theme	Strategic Implication
1	**China's Growing Self-Sufficiency**	China's coal-to-MEG and integrated refinery expansions are progressively reducing import dependency. Coal-to-EG accounted for ~34% of China's EG output in 2024. Exporters who previously relied on the Chinese market must diversify or face volume losses.
2	**Structural Oversupply Risk**	Global capacity utilization has fallen to ~50% by 2024. Unless high-cost capacity is permanently rationalized (as Dow has begun doing), the overhang will continue to depress margins and incentivize price competition.
3	**Trade Tariff Impacts**	U.S.–China tariffs have disrupted established trade flows, forcing U.S. MEG exports to seek alternative markets (India, Indonesia, SE Asia). This adds logistics cost and reduces the net-back for U.S. producers.
4	**Commodity-to-Specialty Pivot**	Dow's strategic shift from MEG to purified EO derivatives (ethanolamines, glycol ethers) signals a broader industry trend: producers with technology and integration capabilities will migrate up the value chain, leaving commodity MEG to the lowest-cost producers.
5	**Shell's Technology Licensing Moat**	Shell's dual role as licensor and producer provides a defensible,

#	Theme	Strategic Implication
		cycle-resilient revenue stream. With ~50% of global EO made on Shell catalysts and 40% in Shell-designed plants, the company earns income regardless of MEG price direction.
6	**SABIC's Feedstock Cost Advantage**	Saudi ethane-based production remains the global marginal cost leader. SABIC can weather prolonged downturns that would be unprofitable for naphtha-based or coal-based producers. The 700 KTA Jubail expansion reinforces this position.
7	**Sustainability & ESG Pressures**	Growing scrutiny of carbon intensity in petrochemical production. SABIC's Jubail CCU plant (500 KTA CO_2 capture) and the SABIC/Linde/SD decarbonization MoU position the company for a low-carbon EG future. Shell's high-selectivity catalysts reduce ethylene waste and energy consumption per tonne of EO.

7.2 Market Forecast & Outlook

Demand Growth: Global MEG demand is expected to continue growing at approximately **3.2% CAGR** through 2030, driven by polyester fiber demand in emerging Asian markets, continued PET packaging growth, and niche applications such as EV thermal-management fluids and green solvents.

Supply Rationalization: The market requires significant capacity rationalization to restore healthy utilization rates. Dow's Seadrift EG closure is an early indicator; further closures or conversions of high-cost units in China (marginal syngas-based plants), Europe (high-energy-cost naphtha-based units), and potentially other North American sites are expected.

Price Outlook: MEG prices are expected to remain range-bound in the $400–$550/MT corridor (U.S. FOB Houston) through 2025–2026, absent a major supply disruption. Upside catalysts include geopolitical supply disruptions, accelerated demand recovery, or faster-than-expected rationalization. Downside risks include further Chinese capacity additions and trade tariff escalation.

Competitive Dynamics: The market will increasingly bifurcate between:

- **Low-cost commodity producers** (SABIC, advantaged Chinese ethylene-based producers) who compete on scale and feedstock cost
- **Technology and integration players** (Shell, Dow) who extract value through licensing, catalysts, and migration to higher-margin EO derivatives
- **Marginal producers** (high-cost syngas-to-MEG, unintegrated European/Asian units) who face existential margin pressure and potential exit

◆ **Strategic Outlook Summary**

The MEG market is transitioning from a growth-driven expansion phase (2019–2022) to a rationalization and margin-management phase (2023–2028). Winners will be those with structural cost advantages (SABIC), technology moats (Shell), or the strategic flexibility to pivot toward higher-value derivatives (Dow). Producers without these attributes face an increasingly challenging competitive environment.

8. Appendix

8.1 Glossary of Key Terms

Abbreviation	Definition
MEG	Monoethylene Glycol — the primary commercial grade of ethylene glycol ($C_2H_6O_2$)
DEG	Diethylene Glycol — a co-product of MEG production, used in plasticizers and solvents
TEG	Triethylene Glycol — a co-product used in natural gas dehydration
EO	Ethylene Oxide — the direct precursor to MEG; also used in surfactants, ethanolamines, and glycol ethers
EG	Ethylene Glycol — general term encompassing MEG, DEG, and TEG
PET	Polyethylene Terephthalate — a polyester polymer used in bottles, packaging, and fibers
PEF	Polyethylene Furanoate — a bio-based alternative to PET under development
PTA	Purified Terephthalic Acid — the other key feedstock (with MEG) for polyester/PET production
KTA	Kilo Tonnes per Annum — thousands of metric tonnes of annual production capacity
TPA	Tonnes per Annum — annual production volume
MM MT	Million Metric Tonnes
CFR	Cost and Freight — trade term where the seller pays freight to the named port of destination
FOB	Free on Board — trade term where the seller's obligation ends at the port of shipment
CIF	Cost, Insurance, and Freight — trade term where the seller pays freight and insurance to the destination port

Abbreviation	Definition
CAGR	Compound Annual Growth Rate
CCU / CCUS	Carbon Capture and Utilization / Carbon Capture, Utilization, and Storage
CSB	U.S. Chemical Safety and Hazard Investigation Board
HS Catalyst	High Selectivity catalyst — Shell's advanced EO catalyst technology achieving 86%+ initial selectivity
NWE	Northwest Europe — common pricing region for European chemical benchmarks

8.2 Data Sources & Methodology Notes

This report draws upon publicly available data and analysis from the following sources. Where exact figures are unavailable, estimates are derived from triangulation of multiple sources and are clearly marked with "~" or "est." notations.

Source Category	Organizations / Publications
Industry Data Providers	ICIS (Supply & Demand Database, pricing), ChemAnalyst, CCFGroup, Tecnon OrbiChem, Intratec Primary Commodity Prices
Market Research	Grand View Research, Research and Markets (ChemAnalyst MEG report), Statista
Company Disclosures	Dow Inc. (press releases, SEC filings), SABIC (annual reports, press releases), Shell Catalysts & Technologies (product documentation, technology briefs)
Industry Organizations	AFPM (American Fuel & Petrochemical Manufacturers), IPC conference proceedings
Government / Regulatory	U.S. Chemical Safety Board (CSB) investigation reports, China National Bureau of Statistics, China Petroleum and Chemical Industry Federation
Trade Publications	ICIS News, Plastics News / Plasteurope, PUdaily, S&P Global Commodity Insights

Methodology: Capacity and production figures are assembled from multiple industry databases and cross-referenced with company announcements. Price data represents assessed annual ranges based on

published benchmarks and should not be interpreted as transaction-level pricing. Forecast figures represent consensus estimates and the author's analysis based on available data through Q4 2024. All tonnage figures are in metric tonnes unless otherwise noted.

Disclaimer: This report is prepared for informational purposes only and does not constitute investment advice or a recommendation to buy or sell any security or commodity. The data and analysis contained herein are based on publicly available information and industry estimates; actual figures may differ from those reported. The author assumes no liability for decisions made based on this report.

— End of Report —